FleetGuard: Annexure

Supporting results and design details for the Round 1 one-slider. Team RAG, Teliport Season 4, Case Study 1 (Predictive Maintenance, NASA C-MAPSS).

Status key: **BUILT** = implemented and run on FD001. **PLANNED** = design only, not yet implemented.

1. FD001 prototype results (BUILT)

Model	CV RMSE	CV MAE	CV NASA*	Test RMSE	Test MAE	Test NASA
Ridge	15.614	12.042	81,508.1	16.184	13.041	390.200
Random Forest (selected)	**11.737**	7.900	**47,893.3**	**13.398**	**9.939**	**261.450**
LightGBM	11.896	7.821	50,083.5	14.520	11.185	299.863

- **CV:** pooled out-of-fold predictions, 5-fold GroupKFold by engine ID (seed 42), true RUL capped at 125.
- **Test:** last recorded cycle of each of the 100 test engines, scored against the raw true RUL from RUL_FD001.txt. The final models were refit on all 100 train engines after selection.
- **Selection:** Random Forest was chosen on pooled CV RMSE, not on test. On test it improves RMSE by about 17% and NASA score by about 33% over Ridge (lower is better for both).
- *CV NASA is summed over all ~20.6k out-of-fold rows, so it is not comparable with the test NASA score (one prediction per engine, 100 engines). NASA score penalizes predicting too much life more than too little.

2. Prototype setup (BUILT)

Item	Detail
Dataset	NASA C-MAPSS FD001: 100 train engines (20,631 cycles), 100 test engines (13,096 cycles).
Sensors	Kept 14: sensor_2, 3, 4, 7, 8, 9, 11, 12, 13, 14, 15, 17, 20, 21. Dropped 7: sensor_1, 5, 6, 10, 16, 18, 19 (population std ≤ 0.01 on EWMA-smoothed train data).
Cleaning	Missing values filled from past readings only (causal). EWMA smoothing (α = 0.3). Z-score scaling fit on train data only (fold-local during cross-validation).
Features	Rolling mean, std and slope over windows 5, 10 and 30 cycles (causal), plus baseline delta (deviation from the engine's first 10 cycles): 154 feature columns.
Target	RUL = last cycle − current cycle, capped at 125 cycles.
Random Forest	n_estimators = 100, max_depth = 12, min_samples_leaf = 5, max_features = 0.7.
Compute	CPU only, two workers. RF cross-validation 573 s, final fit 139 s, full pipeline 777 s.
Quality	60 automated tests passing.

3. Evidence plots (BUILT)

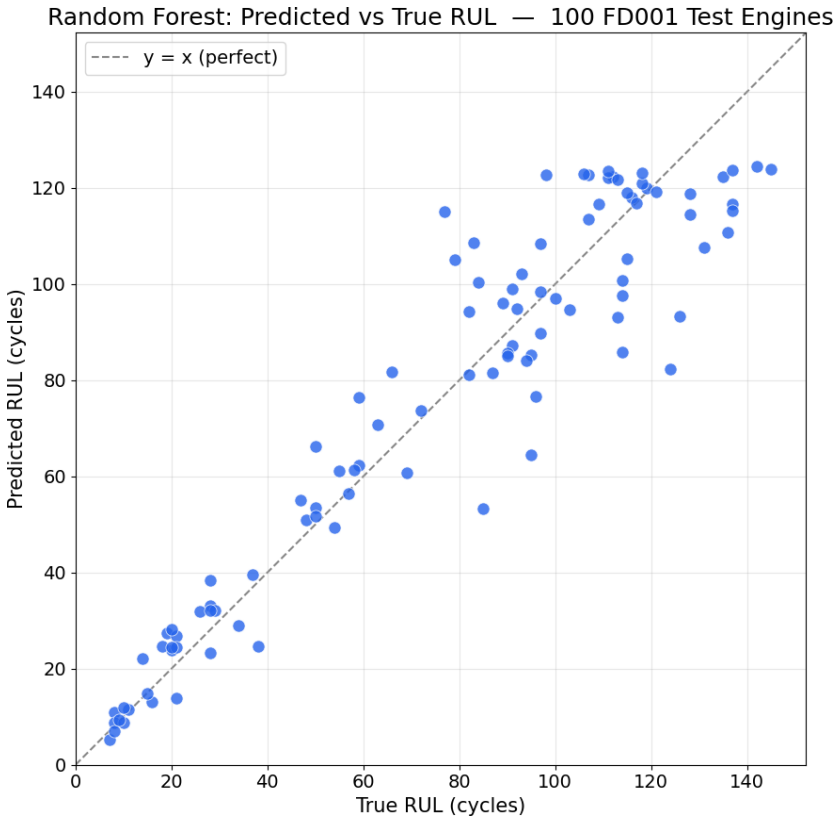


Figure 1. Random Forest, predicted vs true RUL on the 100 FD001 test engines (last cycle of each). Points close to the dashed diagonal are accurate; the scatter is wider at higher RUL.

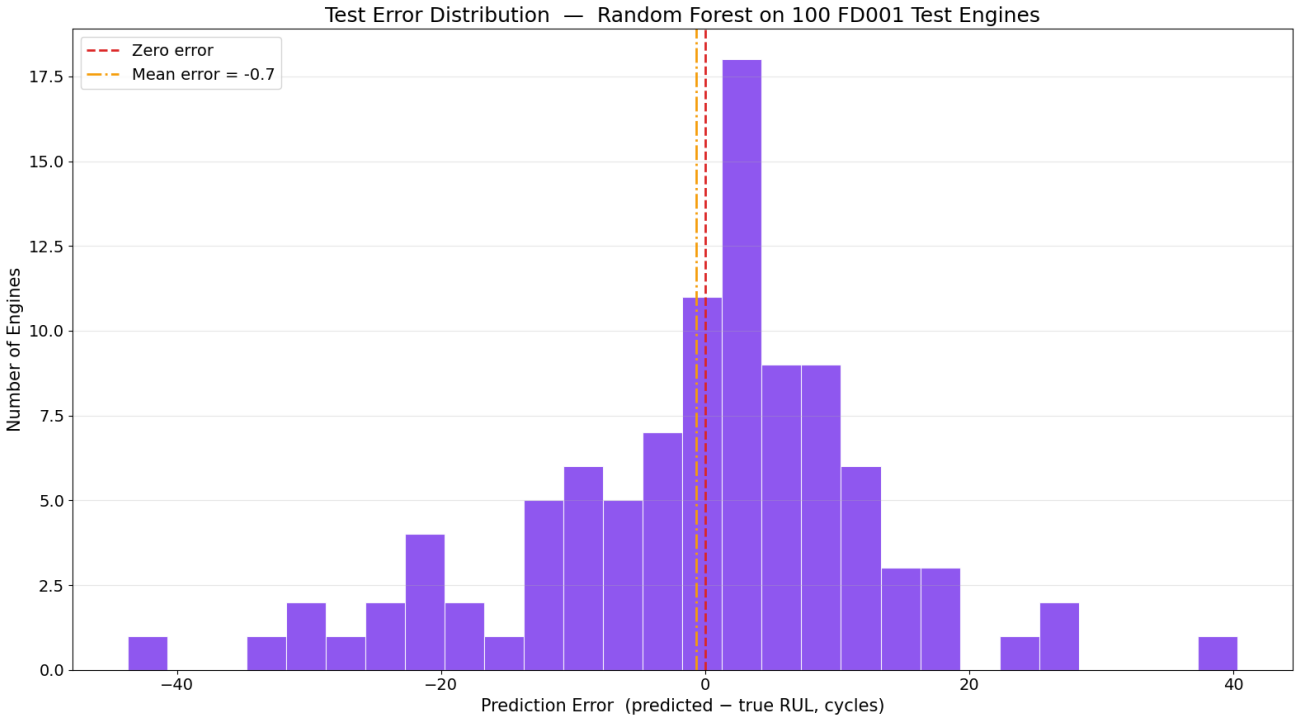


Figure 2. Test error distribution (predicted − true RUL, cycles) for the Random Forest. Mean error is −0.7 cycles; individual errors range from roughly −43 to +40 cycles.

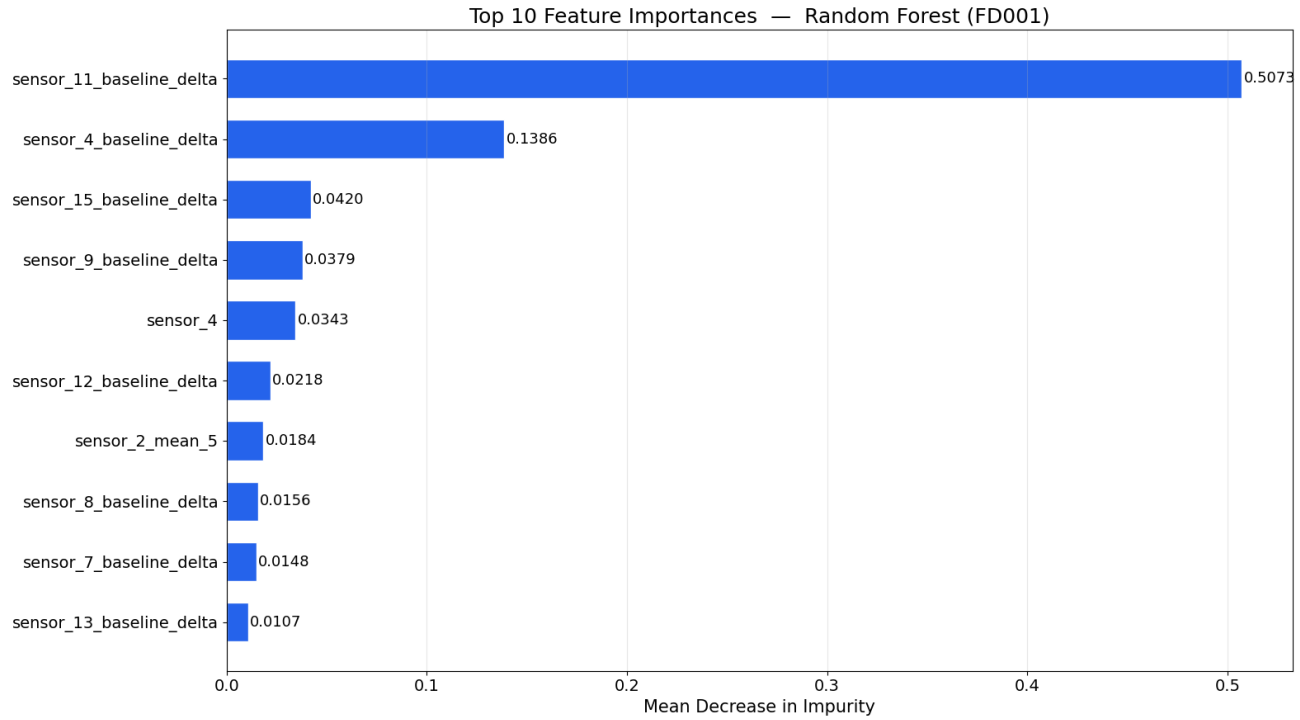


Figure 3. Top 10 Random Forest features (mean decrease in impurity). Deviation from an engine's own healthy baseline dominates: sensor_11 baseline delta alone is 0.507 and sensor_4 baseline delta is 0.139.

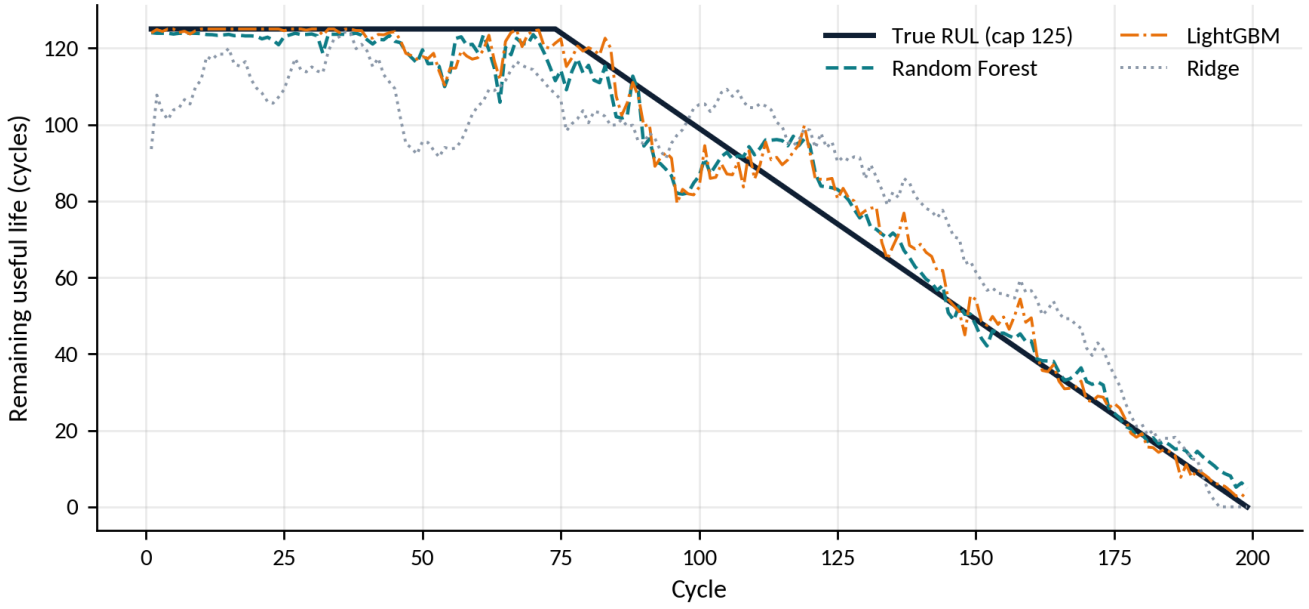


Figure 4. Engine 26 (199 cycles): out-of-fold predictions of all three models against the true RUL (capped at 125), plotted from the exported per-cycle values.

4. Triage agent rules (PLANNED)

Plain-Python, rule-based, no LLM. All thresholds live in a config file.

Severity	Trigger: RUL lower bound (cycles)	Work-order action
CRITICAL	≤ 20	Immediate, top priority
HIGH	≤ 50	Priority, next service window
MEDIUM	≤ 100	Planned, routine slot
NORMAL	> 100	No work order, keep monitoring

- **Inputs per engine:** RUL lower bound (prediction minus k × spread across the forest's trees, k in config), anomaly score, short-term RUL trend, confidence.
- **Anomaly escalation:** a high anomaly score raises severity by one level.
- **Confidence gate:** wide spread or conflicting signals send the case to human review instead of auto-creating an order. Humans handle exceptions only.
- **De-duplication:** no second open order for the same engine unless severity escalates.
- Thresholds are illustrative starting values, to be tuned on validation engines. The lower bound is a design choice; its coverage will be checked on held-out engines.

5. Work order (PLANNED)

Fields: work_order_id, engine_id, subset, severity, predicted_rul, anomaly_score, confidence, top_drifting_sensors, recommended_action, created_at, status. Stored as JSON and in a SQLite table.

6. MLOps loop (PLANNED)

1. New data arrives; a drift check (PSI and KS test on features) compares it with the training reference.
2. Drift above the threshold, or newly arrived labeled data, triggers retraining of a challenger model, tracked in MLflow.
3. The challenger is compared with the current champion on engine-grouped CV; it is promoted in the model registry only if it is better.
4. A Streamlit monitoring view shows fleet RUL and severity, work orders, model versions and drift.

7. Dataset coverage plan

Subset	Operating conditions	Fault modes	Approach	Status
FD001	1	1	Baseline pipeline (as built above)	**BUILT**
FD002	6	1	Cluster the 3 operating settings, normalize sensors per regime	**PLANNED**
FD003	1	2	Same pipeline; check behavior across two fault modes	**PLANNED**
FD004	6	2	Per-regime normalization plus two fault modes	**PLANNED**

8. Known limitations

- Results are for FD001 only; FD002 to FD004 have not been run yet.
- Test metrics come from a single run (seed 42); repeat-seed variance is not reported yet.
- Anomaly model, triage agent, work orders, MLflow tracking, drift check and dashboard are design only at this stage.
- Severity thresholds and the RUL lower bound are untested design choices.